NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

SWD - System Administration

T2-141

Wednesday, 16 July 2025

8:30 AM to 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

QUESTIONS and ANSWERS BOOKLET

TRADE (s): SOFTWARE DEVELOPMENT (SWD)

SUBJECT: SYSTEM ADMINISTRATION

DURATION: 3 Hours

Read carefully the instructions on page 3

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2024-2025

TRADE: SOFTWARE DEVELOPMENT

SUBJECT: SYSTEM ADMINISTRATION

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INSTRUCTIONS TO CANDIDATE:

1. This Exam paper is composed of **three sections (A, B and C).** Follow the instructions given below and answer the indicated questions for a total of **100 marks.**

   Section **A:** Fourteen (14) questions which are **Compulsory**. **55 marks**

   Section **B:** Among the five (5) questions, attempt any three (3) **30 marks**

   Section **C:** Among the two (2) questions, attempt anyone (1). **15 marks**

2. Fill in your actual **names** and **Index number** on the provided space(Cover).
3. Do not remove or tear any page or part of it of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set (English).
5. In section **A**, answer questions in the provided space. In case of multiple choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
   - Blue or Black pen.
   - Geometrical Instruments
   - Silent non-programmable calculators

SECTION A: Attempt all questions **(55 marks)**

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01. Read the questions below and circle the right answer for each: **(3Marks)**

I. Choose correct statement that describes the definition of "computer network":

A. It is a collection of hardware components and computers interconnected by communication channels that do not allow sharing of resources and information.

B. It is a network Architecture in which each computer on the network is either a client or a server network.

C. It is a Powerful computer that provides services to other computers on the network.

D. It is a group of computers connected with each other to communicate and share information and resources like hardware, data, and software.

II. Which of the following best describes the primary purpose of a Peer-to-Peer (P2P) network?

A. Centralized data management

B. Direct communication between devices without a central server

C. Data storage on a central server

D. Voice communication

02. From the list below, identify and categorize each item into the correct column: **(5Marks)**

Networking Hardware or **Network Infrastructure**.

- Firewall and Gateway
- Coaxial cable
- Switch
- Fiber optic cable
- Access Point
- Ethernet cables
- Router
- Antenna Wi-Fi

Networking hardware	Network Infrastructure

03. Read the questions below and circle the right answer for each: **(4Marks)**

I. FROM the options provided, which binary value represents the decimal number 192 in an IP address?

a) 11000000

b) 11100000

c) 10100000

d) 10000000

II. What is the range of private IP addresses in Class C from the options given below?

a) 10.0.0.0 – 10.255.255.255

b) 172.16.0.0 – 172.31.255.255

c) 192.168.0.0 – 192.168.255.255

d) 224.0.0.0 – 239.255.255.255

04. What is the initial step in the network troubleshooting process from the following options? (Circle the correct option) **(3Marks)**

a) Resetting network settings

b) Implementing a solution

c) Replacing network cables

d) Collecting network system information

05. What is the proper sequence of steps to configure a new Forward Lookup Zone in DNS Manager? (Circle the correct sequence) **(4Marks)**

a) Right-click on "Forward Lookup Zones" → Select "New Zone" → Follow the wizard → Specify zone type and name → Choose replication options → Add resource records

b) Right-click on "Forward Lookup Zones" → Select "New Zone" → Specify zone name → Choose replication options → Manage properties

c) Right-click on "Forward Lookup Zones" → Select "New Zone" → Follow the wizard → Specify zone type, name, replication options → Add resource records

d) Right-click on "Forward Lookup Zones" → Select "New Zone" → Choose DNS database location → Specify zone type and name → Manage resource records

06. Read each of the statements below and circle the correct answers: **(5Marks)**

I. Which of the following is used to test local URL accessibility in IIS?

a) Open the website in a web browser using localhost

b) Restart the physical server

c) Remove all site bindings

d) Disable the firewall

II. Which of the following are hosting platforms?

a) Free Hosting

b) Paid Hosting

c) Web Browser

d) Cloud Hosting

07. Identify whether the following statements about network requirements are **TRUE(T)** or **FALSE(F)**: **(4Marks)**

a) A network is a collection of devices connected together to share resources and information……

b) A client-server network has no central control; all devices are equal……

c) The main benefit of networking is to increase data sharing and communication efficiency……

d) A client-server network has no central control; all devices are equal……

08. Circle the characteristic of IPv6 from the options below: **(3Marks)**

a) Uses 32-bit addresses

b) Supports multicast addressing

c) Uses decimal notation

d) Has fewer unique addresses than IPv4

09. How does a private IP address differ from a public IP address? (Circle TWO correct answers). **(4Marks)**

a) A private IP address is used within a local network, while a public IP address is accessible over the internet.

b) A private IP address is not routable on the internet, while a public IP address can be accessed globally.

c) A private IP address is assigned by an ISP, while a public IP address is assigned by a router.

d) A private IP address is used only by servers, while a public IP address is used by personal computers.

10. Fill in the blanks with appropriate terms: **(5Marks)**

a) A _____ is a computer or software that provides services to other computers in a network.

b) _____ is the process of running multiple virtual machines on a single physical server.

c) RAID stands for _____ Array of Independent Disks.

d) _____ is a directory service used by Windows Server for identity and access management.

e) In DHCP, _____ is the period for which an IP address is assigned to a client.

11. You are given a network 192.168.1.0/26 and requested to configure a subnet mask for it. Which of the following subnet masks would correctly configure the network for subnetting? (Circle the correct option). **(4Marks)**

a) 255.255.255.224

b) 255.255.255.192

c) 255.255.255.128

d) 255.255.255.240

12. You need to create a new user account for a department employee in Windows Server 2012 R2. Which of the following steps ensure compliance with organizational policies for password complexity and expiration? (Circle all that apply): **(3Marks)**

a) Enable password complexity requirements in Group Policy settings.

b) Set the password to "Password123" to meet complexity requirements.

c) Configure the account to require a password change at the next logon.

d) Create the user account without setting any password expiration policies.

e) Use a password that includes uppercase letters, lowercase letters, numbers, and special characters.

13. When creating a new virtual machine on VMware ESXi, which of the following steps should be performed first? (Circle the correct answer) **(3Marks)**

a) Power on the virtual machine

b) Access the VMware vSphere Client and connect to the ESXi host

c) Install the guest operating system

d) Configure networking settings

14. User account policies in Windows Server establish the rules and settings for securing and managing user accounts. **(5Marks)**

State whether the following statements about account policies in Windows Server are **TRUE** or **FALSE**.

a) Kerberos policies are only used for local user accounts and do not apply to domain-based authentication.....

b) Logon policies allow administrators to restrict user access based on specific hours or workstations. .....

c) Password policies can enforce a history rule to prevent users from reusing previous passwords. .....

d) Account lockout duration can be configured to automatically unlock accounts after a set period.....

e) Administrators can apply logon scripts through logon policies to automate tasks upon user login.....

Section B: Attempt any three (3) questions **(30 marks)**

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15. Analyze the differences between IPv4 and IPv6 in terms of address structure and efficiency. **(10Marks)**

16. The following steps are used to add users to a group in Windows Server 2012 R2. Analyze the steps below and arrange them in order from step1 to 9. **(10Marks)**

Steps are the following:

Step 1. Press the Windows key + R, type "lusrmgr.msc" and press Enter

Step 2. Click on "Groups" in the left-hand pane

Step 3. In the Properties dialog box, go to the "Members" tab.

Step 4. Log in to the Windows Server 2012 R2 system using an account with administrative privileges.

Step 5. Double-click on the group to which you want to add users.

Step 6. The selected users will now be added to the group.

Step 7. Enter the usernames of the users you want to add to the group.

Step 8. Click the "Add" button to open the Select Users dialog box.

Step 9. Click the "OK" button to close the Select Users dialog box.

17. You are analyzing the performance requirements for a website with backend server-side functionality on a paid hosting platform. What technical aspects would you analyze? **(10Marks)**

18. How would you assess the effectiveness of using Active Directory groups for managing access to resources in a Windows Server environment? Justify your answer. **(10Marks)**

19. A company needs to temporarily disable a user's account due to suspected security concerns. Compare the pros and cons of deactivating the account versus permanently deleting it. **(10Marks)**

Section C: Attempt only one (1) question **(15 marks)**

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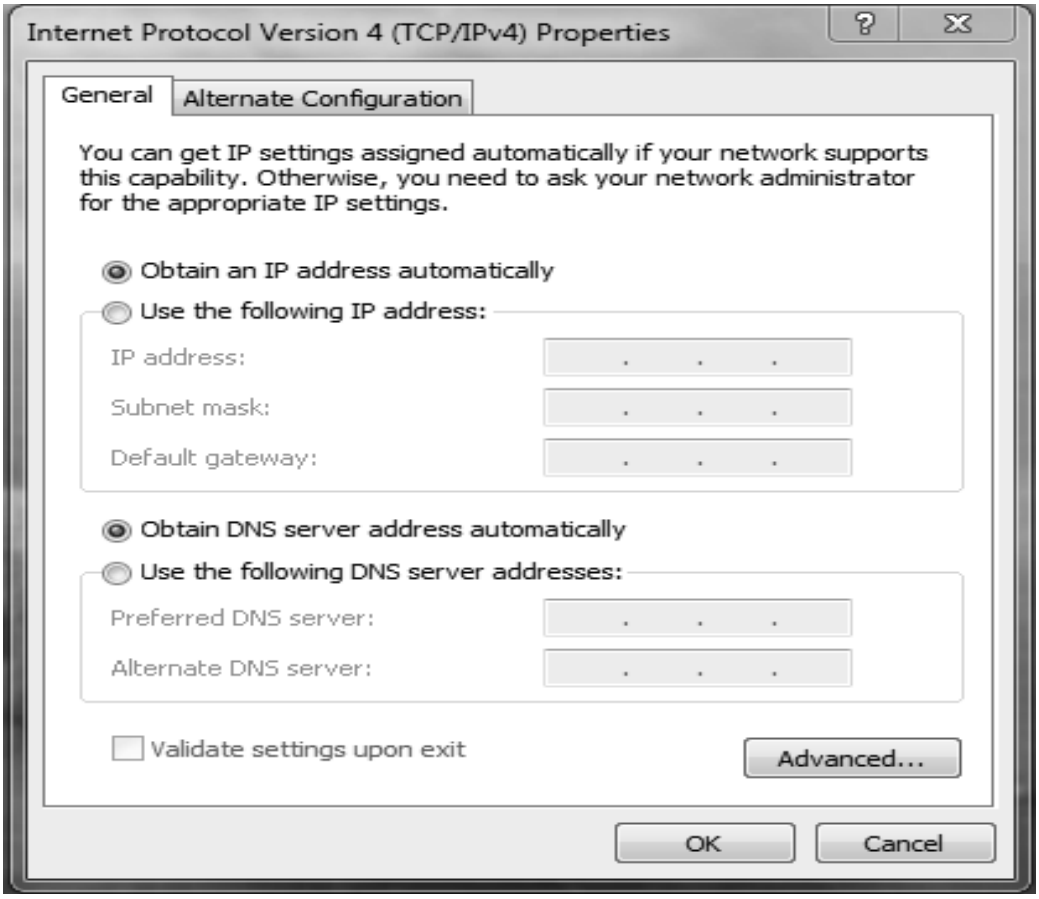
20. Read the questions given below and provide the answers: **(6Marks)**

a) Below is the IPv4 properties window on which the IP address, subnet mask, default gateway and DNS server addresses can be manually configured on a local computer which uses the Windows 7 operating system.



What steps would you follow to get to this window and how would you manually configure those IP addresses?

b) Create a subnetting plan to divide the network 192.168.1.0/24 into four equal subnets and determine the subnet mask. **(9Marks)**

21. Create a plan for testing and verifying the accessibility, speed, and size of a web app after it has been successfully deployed on a Windows Server using IIS. **(15Marks)**

END

RESERVED SPACE FOR SECTION B AND C ANSWERS

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